tear open the side of a mountain to attach the mouth of a tunnel, we must tear open the fabric of space to attach the mouth of a wormhole.[12] No one knows whether such tears in space are allowed by the laws of physics. Work with which I've been involved in string theory (see this page) has shown that certain kinds of spatial tears are possible, but so far we have no idea whether these rips might be relevant to the creation of wormholes. The bottom line is that intentional acquisition of a macroscopic wormhole is a fantasy that, at best, is a *very* long way from being realized.

Morever, even if we somehow managed to get our hands on a macroscopic wormhole, we wouldn't be done; we'd still face a couple of significant obstacles. First, in the 1960s, Wheeler and Robert Fuller showed, using the equations of general relativity, that wormholes are unstable. Their walls tend to collapse inward in a fraction of a second, which eliminates their utility for any kind of travel. More recently, though, physicists (including Thorne and Morris, and also Matt Visser) have found a potential way around the collapse problem. If the wormhole is not empty, but instead contains material—so-called *exotic matter*—that can exert an outward push on its walls, then it might be possible to keep the wormhole open and stable. Although similar in its effect to a cosmological constant, exotic matter would generate outward-pushing repulsive gravity by virtue of having negative energy (not just the negative pressure characteristic of a cosmological constant[13]). Under highly specialized conditions, quantum mechanics allows for negative energy,[14] but it would be a monumental challenge to generate enough exotic matter to hold a macroscopic wormhole open. (For example, Visser has calculated that the amount of negative energy needed to keep open a one-meter-wide wormhole is roughly equal in magnitude to the total energy produced by the sun over about 10 billion years.[15])

Second, even if we somehow found or created a macroscopic wormhole, and even if we somehow were able to buttress its walls against immediate collapse, and even if we were able to induce a time difference between the wormhole mouths (say, by flying one mouth around at high speed), there would remain another hurdle to acquiring a time machine. A number of physicists, including Stephen Hawking, have raised the possibility that vacuum fluctuations—the jitters arising from the quantum uncertainty experienced by all fields, even in empty space,